

Soil & Air Environmental Telemetry Studio

Comprehensive Technical Report: Digital Signal Processing, VPD Analytics & Container
Soil-Hydrology Physics

Experiment: Indoor Unplanted Soil Pot	Hardware: ESP32 Dual Sensor Suite
Location: Near Window (Natural Airflow)	Telemetry URL: soil-air-dashboard.pages.dev
Direct Sun Timing: Incident After 1:30 PM	Author / Research: Apurba Maity & Ciel

1. Soil Moisture DSP Filter Pipeline & Mathematical Derivations

The raw capacitive analog soil moisture probe (ADC1 GPIO34) is inherently prone to three physical distortions: high-frequency Wi-Fi transmission RF burst noise, non-linear electrostatic fringe fields, and thermal dielectric drift. To resolve these anomalies, a four-stage Digital Signal Processing (DSP) pipeline was designed:

Stage 1: Median Filter (N=5)	Equation: $x_{med}[n] = \text{median}(x[n-2], \dots, x[n+2])$ Function: Strips non-Gaussian impulse spikes and radio-frequency noise without phase shift.
Stage 2: Power-Law Calibration	Equation: $\theta_{cal} = ((2040 - ADC_{med}) / (2040 - 1080))^{1.15} \times 100\%$ Function: Linearizes non-linear fringe capacitance near dry/wet saturation thresholds.
Stage 3: Temperature Compensation	Equation: $\theta_{comp} = \theta_{cal} - 0.35 \times (T_{soil} - 25.0^{\circ}\text{C})$ Function: Corrects the temperature dependence of water permittivity ($\partial\epsilon/\partial T \approx -0.35\%/^{\circ}\text{C}$).
Stage 4: 1st-Order IIR Low-Pass	Equation: $y[n] = 0.25 \times x[n] + 0.75 \times y[n-1]$ Function: Exponential moving smoothing with a cutoff frequency below environmental noise.

Why the Filtered Value is Always Lower Than Raw Percentage Readings

You observed that the Emerald-Green DSP filtered curve is consistently **5% to 8% lower** than the raw capacitive/resistive values. This is mathematically guaranteed by two physical corrections:

- Non-linear Power-Law Curvature ($\alpha = 1.15$):** Because the exponent $1.15 > 1.0$, any normalized ratio $u \in (0, 1)$ satisfies $u^{1.15} < u$. For instance, an uncalibrated linear ratio of 0.70 (70%) becomes $0.70^{1.15} = 0.663$ (66.3%), removing overestimation in the mid-range.
- Soil Temperature Dielectric Offset ($\Delta T > 0$):** Water dielectric permittivity naturally decreases as temperature increases. Because your room soil temperature is currently **29.5°C to 32.0°C** (higher than the 25.0°C calibration baseline), the term $-0.35 \times (31.0 - 25.0)$ subtracts an additional **2.1%** to compensate for false thermal capacitance.

2. Vapor Pressure Deficit (VPD) & Atmospheric Drying Demand

Vapor Pressure Deficit (VPD in kPa) measures the difference between the moisture currently in the air and the maximum moisture the air can hold when saturated. It is the absolute thermodynamic driving force behind soil surface evaporation and plant transpiration.

1. Saturation Vapor Pressure (VP_{sat}):	$VP_{sat} = 0.61078 \times \exp((17.27 \times T_{air}) / (T_{air} + 237.3))$ [kPa]
2. Actual Vapor Pressure (VP_{act}):	$VP_{act} = VP_{sat} \times (RH / 100)$ [kPa]
3. Vapor Pressure Deficit (VPD):	$VPD = VP_{sat} - VP_{act} = VP_{sat} \times (1 - RH / 100)$ [kPa]

VPD Range	Microclimate State	Physical / Agronomic Meaning
< 0.4 kPa	Low Evaporation	Air is nearly saturated; stagnant evaporation, high condensation risk.
0.4 – 1.6 kPa	Optimal Zone	Healthy, balanced transpiration and steady soil surface drying.
> 1.6 kPa	High Drying Stress	Dry, hot air rapidly sucks moisture from topsoil (occurs during 1:30 PM sun).

3. Unplanted Container Soil Physics: The 1:30 PM Direct Sun & Nocturnal Capillary Upflow

In an unplanted indoor soil container, soil moisture does not decay monotonically. Instead, your empirical telemetry records three distinct daily physical regimes:

1. Morning Baseline (06:00 – 13:30)	Condition: Diffused room light, ambient window ventilation. Dynamics: Air temp $\approx 30.5^{\circ}\text{C}$, RH $\approx 76\%$. Soil undergoes gentle, steady baseline surface evaporation ($d\theta/dt \approx -0.2\%/hr$).
2. Afternoon Solar Pulse (13:30 – 17:30)	Condition: Direct solar infrared radiation strikes container. Dynamics: Soil temp rapidly jumps to 32°C – 39°C . High thermal kinetic energy causes rapid surface evaporation flash ($d\theta/dt \approx -1.5\%$ to $-3.0\%/hr$, sharp amber valley).
3. Nocturnal Recharge (20:00 – 05:00)	Condition: No sun, air cools, window humidity reaches 80–84% RH. Dynamics: Topsoil was dried by afternoon sun (high matric suction). Moisture from damp subsoil moves UPWARD by capillary action, while humid vapor adsorbs into pores ($d\theta/dt \approx +0.8\%$ to $+1.5\%/hr$, emerald peak).

4. Understanding the Research Lab Studio Graphs

Soil Thermal Hysteresis (T_{soil} vs T_{air})	Plots a 2D phase portrait of Soil Temp vs Air Temp. The enclosed loop area measures volumetric heat capacity. Because water has a high specific heat ($4.184\text{ J/g}^{\circ}\text{C}$), a wetter pot produces a wider, fatter loop; as the pot completely dries, the loop collapses into a narrow line.
Moisture Dynamics Rate ($d\theta/dt$ in $\%/hr$)	Computed using a 5-point rolling central-difference derivative. Amber negative bars show peak afternoon solar drying; Emerald positive bars show nighttime capillary rise and vapor adsorption.
Dew Point Margin ($\Delta T_{\text{dew}} = T_{\text{air}} - T_{\text{dew}}$)	Measures proximity to 100% saturation. When $\Delta T_{\text{dew}} > 5^{\circ}\text{C}$, evaporation occurs freely. When ΔT_{dew} approaches 1.5°C (red dashed line), ambient air can no longer accept vapor, and adsorption/condensation occurs.
Dielectric vs Conductive Discrepancy	Tracks $\theta_{\text{Cap}} - \theta_{\text{Res}}$. Isolates dissolved fertilizer and ion concentration effects from true volumetric dipole water content.

5. Gravimetric Scale Weight Cross-Validation Protocol

To calibrate the electronic sensor readings against physical mass, total container weight $W(t)$ is logged on a digital scale in grams. The relationship follows mass conservation:

Evaporation Mass Loss Rate $[\text{g/hr}] = \Delta W / \Delta t = (W(t_1) - W(t_2)) / (t_2 - t_1)$

Telemetry Rate Proportionality: $d\theta/dt \approx -k \times (\Delta W / \Delta t)$

Once your scale weight logs are entered, the proportionality constant k will calibrate the dashboard to display **water loss in grams/hour** directly in real time.